

EXACT POWERS FOR THE LINEAR CASE OF WILKS' LIKELIHOOD RATIO CRITERION

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1. Summary

The noncentral linear distribution of Wilks' likelihood ratio criterion is expressed in a general form, and an algorithm is developed which calculates exact percentiles and powers of the criterion. The algorithm is valid for all parameter values except the case where m , the degrees of freedom for hypothesis, and p , the number of variates, are both odd, and is otherwise limited only by computer precision. The accuracy of certain published asymptotic powers is examined and a selection of results with varying parameters is included.

2. Introduction

Let the random matrices $A(p \times p)$ and $B(p \times p)$ be independently distributed according to $W(I, n)$ and $W(I, m, \lambda^2)$ respectively, where $W(I, m, \lambda^2)$ denotes a Wishart distribution with covariance matrix I , m degrees of freedom and linear noncentrality parameter λ^2 . If $A = CLC'$, $A+B = CC'$ and C is a lower triangular matrix, then $L = C^{-1}AC^{-1}$ has a noncentral linear multivariate Beta type I distribution (Troskie (1966)) and

$$(2.1) |L| = |A|/|A+B|$$

is Wilks' (1932) likelihood ratio criterion for

- (i) testing linear hypotheses about regression coefficients,
- (ii) testing equality of means of several normal distributions with common covariance matrix, and
- (iii) testing the multivariate general linear hypothesis in MANOVA.

In the central case, or with appropriate changes in the noncentrality parameter of B , $|L|$ is also closely related to test criteria for

- (iv) testing independence between a p -set and a q -set of variates in a $(p+q)$ -variate normal population, and
- (v) testing equality of the covariance matrices of two p -variate normal populations.

We shall also write $|L|$ as $U_{p,m,n}$, and note that A and B represent the sums of pro-

duct matrices for error and hypothesis respectively.

The exact null hypothesis distribution of $U_{p,m,n}$, i.e. when B is also a central Wishart distribution, was found by Wilks (1935) for values of p and $m \leq 4$. Later work by Bartlett (1938), Wald and Brookner (1941), Rao (1948) and Box (1949) developed asymptotic approximations to logarithmic functions of $U_{p,m,n}$ under the null hypothesis. Using the method of convolutions, Schatzoff (1966) derived the central distribution of $U_{p,m,n}$ for either p or m even, and developed an algorithm to tabulate correction factors for converting χ^2 percentiles to exact percentiles of a logarithmic transformation of $U_{p,m,n}$, for selected values of p , m and n . Pillai and Gupta (1969) derived explicit finite series expressions for $U_{p,m,n}$ with m even and $p=3,4,5$ and 6 , also extending Schatzoff's tables, while Lee (1972) simplified the form of the distribution for both p and m odd, further extending the tables to include values for all $p \leq m \leq 20$ and $pm \leq 144$ with omission when p or m is odd and greater than 10.

Anderson and Girshick (1944) defined the linear and planar cases of the non-null distribution, for which various authors have since supplied approximations and limited tabulations of percentage points and powers. Those providing actual numerical values include Roy (1960), Ito (1962), Posten and Bargmann (1964), Mikhail (1965), Pillai and Jayachandran (1967), De Waal (1968), Lee (1971a) and Lee (1971b). Gupta (1971) used Kshirsagar's (1961) derivation of the noncentral distribution of L in the linear case to find explicit expressions for $U_{p,m,n}$ with $p=2,3,4$ and 5 . Money (1972) extended these results to further test criteria for tests (i) to (v) above, in both the real and complex cases.

In this paper we examine the noncentral form of the various statistics mentioned and derive an algorithm which provides exact values of the matrices of coefficients needed to determine exact percentiles. This further enables us to calculate exact powers of the tests.

3. Distribution of $|L|$ in the noncentral linear case

The noncentral Beta type 1 density can be written as

$$(3.1) \beta_1(x_1; \lambda^2, m, n) = (B(\frac{1}{2}m, \frac{1}{2}n))^{-1} x_1^{\frac{1}{2}(n-2)} (1-x_1)^{\frac{1}{2}(m-2)} \\ \exp(-\frac{1}{2}\lambda^2) \sum_{j=0}^{\infty} (\frac{1}{2}(m+n))_j (\frac{1}{2}\lambda^2(1-x_1))^j / ((\frac{1}{2}m)_j j!) \quad 0 \leq x_1 \leq 1$$

where $(a)_j = a(a+1) \dots (a+j-1) = \Gamma(a+j)/\Gamma(a)$ and

$$B(\frac{1}{2}m, \frac{1}{2}n) = \Gamma(\frac{1}{2}m)\Gamma(\frac{1}{2}n)/\Gamma(\frac{1}{2}m + \frac{1}{2}n)$$

while the central Beta type 1 density is

$$(3.2) \beta_1(x_1; m, n) = (B(\frac{1}{2}m, \frac{1}{2}n))^{-1} x_1^{\frac{1}{2}(n-2)} (1-x_1)^{\frac{1}{2}(m-2)} \quad 0 \leq x_1 \leq 1$$

The following theorems are extensions of those given by Anderson (1958) for the central distribution:

Theorem 3.1 Kshirsagar (1961), Gupta (1971)

In the noncentral linear case the statistic $U_{p,m,n}$ is distributed as $\Pi_{i=1}^p X_i$ where the X_i ($i=1,2,\dots,p$) are independent, X_1 is distributed as (3.1) and X_i ($i=2,\dots,p$) are distributed as $\beta_1(x_i; m, n-i+1)$.

Theorem 3.2 Gupta (1971)

In the noncentral linear case $U_{2r,m,n}$ is distributed as $X_1(\Pi_{i=1}^{r-1} Y_i^2)X_{2r}$ where X_1 and X_{2r} are independently distributed as (3.1) and $\beta_1(x_{2r}, m, n-p+1)$ respectively, and the Y_i ($i=1,2,\dots,r-1$) are independently distributed as $\beta_1(y_i; 2m, 2(n-2i))$. $U_{2s+1,m,n}$ is distributed as $X_1\Pi_{i=1}^s Y_i^2$ where X_1 is independently distributed as (3.1) and the Y_i ($i=1,2,\dots,s$) are independently distributed as $\beta_1(y_i; 2m, 2(n-2i))$.

When m is even we can expand powers of $(1-x_i)$ binomially in a finite series, and letting $y_i = -\log x_i$ ($i=1,2,\dots,p$), we have

$$(3.3) \quad f(y_1) = e^{-\frac{1}{2}\lambda^2} (B(\frac{1}{2}m, \frac{1}{2}n))^{-1} \sum_{j=0}^{\infty} a_j \sum_{k=0}^{b+j} (-1)^k \binom{b+j}{k} \cdot \exp(-\frac{1}{2}(n+2k)y_1)$$

$$y_1 \geq 0$$

where

$$a_j = (\frac{1}{2}(m+n))_j (\lambda^2/2)^j / ((\frac{1}{2}m)_j j!), \text{ and}$$

$$b = \frac{1}{2}(m-2), \quad \text{and}$$

$$(3.4) \quad f(y_i) = (B(\frac{1}{2}m, \frac{1}{2}(n-i+1)))^{-1} \sum_{\ell=0}^b (-1)^\ell \binom{b}{\ell} \cdot \exp(-\frac{1}{2}(n-i+2\ell+1)y_i)$$

$$y_i \geq 0; i=2,\dots,p.$$

Gupta (1971) showed that the density function of $Z_i = X_{2i}X_{2i+1}$ is given by

$$(2B(m, n-2i))^{-1} z_i^{\frac{1}{2}} (n-2i-2)(1-z_i^{\frac{1}{2}})^{m-1}$$

and if $Y'_i = -\log X_i$ then

$$(3.5) \quad f(y'_i) = (2B(m, n-2i))^{-1} \sum_{\ell=0}^{m-1} (-1)^\ell \binom{m-1}{\ell} \cdot \exp(-\frac{1}{2}(n+\ell-2i)y'_i) \quad y'_i \geq 0.$$

Since

$$-\log U_{p,m,n} = \sum_{i=1}^p (-\log X_i) = \sum_{i=1}^p Y_i$$

we may use the method of successive convolutions to find the density of $U_{p,m,n}$, noting

Theorem 3.2 and the following Lemma.

Lemma 3.1 Schatzoff (1966)

$$\text{If } V_1 \sim v_1^k e^{sv_1} \quad k > 0 \text{ and integral}$$

$$V_2 \sim e^{tv_2}$$

$$(3.6) \text{ then } V = (V_1 + V_2) \sim e^{tv} v^{k+1} / (k+1) \quad s = t$$

$$(3.7) \quad \sim e^{sv} \left\{ \sum_{r=1}^{k+1} (-1)^{r+1} \frac{k!}{(k-r+1)!} \frac{v^{k-r+1}}{(s-t)^r} \right\} \\ + e^{tv} \left(\frac{1}{t-s} \right)^{k+1} k! \quad s \neq t$$

Gupta (1971) has in this way derived explicit expressions for the density and cumulative distribution function (c.d.f.) of $U_{p,m,n}$ for m even and $p=2,3,4$ and 5.

4. A general form for $U_{p,m,n}$

With successive convolutions the densities and c.d.f.'s of $U_{p,m,n}$ become increasingly unmanageable. Careful examination, however, reveals the occurrence of a pattern which prompts the following theorem:

Theorem 4.1

For any p , and m even, the density of $Y_p = -\log U_{p,m,n}$ may be expressed in a general form as an infinite series involving integral powers of y_p and of $e^{-y_p/2}$;

$$(4.1) \quad f(y_p) = K_p^{-1} e^{-\frac{1}{2}\lambda^2} \sum_{j=0}^{\infty} a_j \sum_{r=1}^s \sum_{c=c_c}^{c_f} d_{jrc} e^{-\frac{1}{2}cy_p} y_p^{r-1}$$

where

$$(4.2) \quad K_p = B\left(\frac{1}{2}m, \frac{1}{2}n\right) \prod_{i=1}^{s-1} (2B(m, n-2i)) \quad p \text{ odd} \\ = B\left(\frac{1}{2}m, \frac{1}{2}n\right) / \left(\prod_{i=1}^{s-2} (2B(m, n-2i)) \right) B\left(\frac{1}{2}m, \frac{1}{2}(n-p+1)\right) \quad p \text{ even} \\ s = \frac{1}{2}(p+1) \quad p \text{ odd} \\ = \frac{1}{2}(p+2) \quad p \text{ even}$$

$$c_c = n-2s+2r \text{ and } c_f = m+n+2j-2$$

Schatzoff (1966) gave a general form for the central density while Gupta (1971) suggested one for the noncentral linear case but did not indicate explicitly how values of the constants could be found. If the coefficients d_{jrc} can be determined for any p , m and n , the density and c.d.f. of $U_{p,m,n}$ are specified. Unlike Schatzoff's central form, this form involves an infinite series expansion in j , which has to be truncated. Truncation error can, however, be controlled as the series is always convergent.

5. Calculation of the coefficients d_{jrc} of $U_{p,m,n}$

The matrix of coefficients $(d_{jrc})_p$ for $p = 1$ is given by

$$(5.1) \quad d_{jlc} = (-1)^{(c-n)/2} \binom{(m-2+2j)/2}{(c-n)/2} \quad (c-n) \text{ even}$$

$$= 0 \quad (c-n) \text{ odd}$$

The matrix of d_{jrc} for the required p may be found by a series of "double" convolutions (using (3.5)), and, for even p , a final "single" convolution (using (3.4)). We now show how the general form of $f(Y_p)$ is affected by convolution.

Theorem 5.1

Let the general form of the j -th term of $f(Y_p)$ be

$$(5.2) \quad Y_p(j) = J_p \sum_{r=1}^s \sum_{c=c_c}^{c_f} d_{jrc} e^{-\frac{1}{2}cy} y_p^{r-1}$$

$$(5.3) \quad \text{where } J_p = K_p^{-1} e^{-\frac{1}{2}\lambda^2} a_j \text{ and}$$

s, c_c, c_f, K_p and a_j are as defined in (4.2).

Then the general form of $Y_{p+1}(j)$ and $Y_{p+2}(j)$, after convolution with (3.4) and (3.5) respectively, is given by

$$(5.4) \quad Y_{p+1}(j) = J_{p+1} \sum_{r=1}^s \sum_{c=c_c}^{c_f} (G_1 + G_2 + G_3)$$

where

$$(5.5) \quad G_1 = (-1)^{\ell} \binom{b^*}{\ell} d_{jrc} e^{-\frac{1}{2}cy_{p+i}} y_{p+i}^r,$$

$$(5.6) \quad G_2 = \sum_{\substack{\ell=0 \\ \ell \neq \ell^*}}^{b^*} (-1)^{\ell+1} \binom{b^*}{\ell} d_{jrc} \sum_{r^*=1}^r \left(\frac{2}{c-n^*} \right)^{r^*} \frac{(r-1)!}{(r-r^*)!} \cdot e^{-\frac{1}{2}cy_{p+i}} y_{p+i}^{r-r^*},$$

$$(5.7) \quad G_3 = \sum_{\ell \neq \ell^*}^{b^*} (-1)^{\ell} \binom{b^*}{\ell} d_{jrc} \left(\frac{2}{c-n^*} \right)^r (r-1)! e^{-\frac{1}{2}n^*y_{p+i}}$$

and for $i = 1$

or $i = 2$

$$(5.8) \quad J_{p+i} = J_p(B(\frac{1}{2}m, \frac{1}{2}(n-p)))^{-1} \quad \text{or} \quad J_p(2B(m, n-2s))^{-1}$$

$$\ell^* = \frac{1}{2}(c+p-n)$$

$$\text{or} \quad c+2s-n$$

$$b^* = \frac{1}{2}(m-2)$$

$$\text{or} \quad m-1$$

$$n^* = n+2\ell-p$$

$$\text{or} \quad n+\ell-2s$$

and

$$t = 1 \text{ for } \ell^* \text{ integral}$$

$$= 0 \text{ for } \ell^* \text{ non-integral}$$

The new matrix of coefficients $(d'_{jrc})_{p+i}$ is the union of sets of contributions from each of the G_k ($k=1,2,3$).

From G_1 , for $r=2, \dots, s$ and $c = c_c, \dots, c_f$, d'_{jrc} receives

$$(5.9) \quad d'(1) = t(-1)^{\ell^*} \binom{b^*}{\ell^*} d_{j(r-1)c}/(r-1)$$

From G_2 , for $r=1, \dots, s$; $c = c_c, \dots, c_f$ and $q=1, \dots, r$, d'_{jqc} receives

$$(5.10) \quad d'(2) = \frac{(r-1)!}{(q-1)!} d_{jrc} \sum_{\substack{\ell=0 \\ \ell \neq \ell^*}}^{\ell^*} (-1)^{\ell+1} \binom{b^*}{\ell} \left(\frac{2}{c-n^*}\right)^{r-q+1}$$

From G_3 , for $c = (c_c-i), \dots, (m+n-2s-1)$, d'_{jlc} receives

$$(5.11) \quad d'(3) = t \sum_{r=1}^s \sum_{\substack{g=c_c \\ g \neq c}}^{c_f} (-1)^{\ell^*} \binom{b^*}{\ell^*} \left(\frac{2}{g-c}\right)^r (r-1)! d_{jrg}$$

The convolution procedure therefore determines the coefficients necessary to specify values of the density and c.d.f., which are given in the same general form below.

Theorem 5.2

The density and c.d.f. of $U_{p,m,n}$ are given by $f(u)$ and $F(u)$ respectively where

$$(5.12) \quad f(u) = K_p^{-1} e^{-\frac{1}{2}\lambda^2} \sum_{j=0}^{\infty} a_j \sum_{r=1}^s \sum_{c=c_c}^{c_f} d_{jrc} \cdot u^{\frac{1}{2}(c-2)} (-\log u)^{r-1} \quad 0 \leq u \leq 1 \text{ and}$$

$$(5.13) \quad F(u) = K_p^{-1} e^{-\frac{1}{2}\lambda^2} \sum_{j=0}^{\infty} a_j \sum_{r=1}^s \sum_{c=c_c}^{c_f} (r-1)! \cdot u^{\frac{1}{2}c} d_{jrc} \sum_{r^*=1}^r \frac{(-\log u)^{r-r^*}}{(r-r^*)!} \left(\frac{2}{c}\right)^{r^*} \quad 0 \leq u \leq 1$$

and

s, c, c_p, K_p and a_j are as defined in (4.2).

The convolution process obviously involves computation and storage of a large number of coefficients. Careful examination of (5.9), (5.10) and (5.11) enable this to be streamlined by noting.

Theorem 5.3

At any stage of convolution, only elements d_{jrc} with the following subscripts may be nonzero:

$r = 1$ and $c = ((m+n-2+2i), i=0,1,\dots,j), \quad \text{all } j$

or $c = ((n-2s+2), \dots, (m+n-2)), \quad \text{all } j$

and

$r > 1$ and $c = ((n-2s+2r), \dots, (m+n-2r)), \quad \text{all } j.$

It follows that all d_{jrc} are zero for $r > \frac{1}{2}m$.

6. Comparison of exact and asymptotic results

In this section we examine the accuracy of some published asymptotic percentiles and powers of $|L|$ in the linear case.

In Table 1 we compare results derived by the algorithm with a comparison by Gupta (1971) of his exact percentage points (calculated from explicit expressions) and a Posten and Bargmann approximation. This confirms Gupta's statement that the Posten and Bargmann approximation is not so good for small $t = n + \frac{1}{2}(m-p-1)$ and large λ^2 . It also shows that for n (and t) large, the approximation is even better than Gupta's "exact" calculation. This is probably due to Gupta's infinite series being truncated after too few terms — more terms are needed as λ^2 increases, and the greatest discrepancies occur when λ^2 is largest (i.e. 4,0).

J. Roy (1960) obtained an asymptotic expression for the distribution of $-n \log |L|$ to the order n^{-1} and tabulated approximate powers for $n = 200$. These were extended by Ito (1962), who compared the approximate powers with those of the Lawley-Hotelling T_0^2 test. Table 2 compares Ito's results with those derived by our method and shows that for $\lambda^2 > 2$, Ito's approximation underestimates the exact power by an amount which increases with λ^2 and decreases with p . The exact powers in Table 2 are all equal to or greater than Ito's approximate powers for the T_0^2 test.

Mikhail (1965) used a Pearson type III approximation to the moments of $(n(1-U)/mU)$ for $p=2$, and compared it with an exact power calculation. He used noncentrality parameter $\Lambda = \Sigma \lambda_1^2$ after showing that the first two moments of his statistic did not depend on any other function of the λ_1^2 , and concluded that "the approximation is evidently excellent for the larger values of N ; for the smaller values of N

it becomes poor for large Λ ." This conclusion is based on an inaccurate calculation of the exact power. His statement that "Ito's approximation is no better than a simple limit using the Patnaik approximation" is also invalid (see Table 3). Mikhail's approximation consistently overestimates the true power by an amount which increases with Λ and decreases with n . This is shown in Table 4, together with an "adjusted" power obtained by subtracting $\Lambda^{3/2}/(4n)$ from Mikhail's approximation.

Posten and Bargmann (1964) have developed approximations of order q^{-1} and q^{-2} to $-q \log U_{p,m,n}$ for the linear and planar cases, where $q = n + \frac{1}{2}(m-p-1)$. These appear to give very satisfactory results (see Table 1). de Waal (1968) derived a χ^2 approximation of order n^{-1} to $-q \log U_{p,m,n}$. Table 5 compares this with J. Roy's (1965) Gamma approximations and our exact results. The χ^2 approximation appears to be better for larger λ . Both Sugiura and Fujikoshi (1969) and Lee (1971b) found asymptotic expansions of $|L|$ for the general noncentral case to the order of q^{-2} . Lee (1971b) tabulated approximate powers of $|L|$ for $p=2,3$ and 4, various values of m and n , and combinations of up to four non-zero values of λ . These provide good approximations to the exact results.

7. Accuracy, convergence and range of results

Exact percentiles and powers of $|L|$ can be found using (5.13) for any values of p , even m, n, λ and α . In addition, Das Gupta and Perlman (1973) have shown that the powers of $U_{p,m,n}$ and $U_{m,p,m+n-p}$ are equal, and the method can thus be used if either p or m is even. For both p and m odd, an extension of Lee's (1972) technique for the central case, or interpolation, might be used.

The main problem (outlined by Schatzoff (1966)) in the use of exact formulae is the severe loss of significant figures through heavy cancellation as p, m and n increase. For example, the central cumulative frequency distribution of $U_{5,10,10}$ at $u=1$, given by

$$(7.1) K_p^{-1} \sum_{r=1}^s \sum_{c=c_c}^{c_f} (r-1)! (2/c)^r d_{jrc} = K_p^{-1} \Sigma_r \Sigma_c Q_{rc}$$

has

$$\max_{r,c} Q_{rc} \text{ of order } 10^5 \text{ but}$$

r, c

$$\Sigma_r \Sigma_c Q_{rc} = K_p \text{ of order } 10^{-12}.$$

The position is not quite as severe with lower values of u , which aids computation of lower percentiles with small n , but is aggravated by increasing j in the noncentral case. Multiple precision facilities are therefore essential for exact percentiles involving combinations of large parameters.

For the lower values of λ^2 , convergence of the infinite series does not present a problem, and as previously stated, truncation error may be controlled. With larger λ^2 however, loss of precision may occur before convergence has started or progressed sufficiently, and a compromise must be reached between the two errors. The problems of convergence and precision generally act in concert, so that although one or two of the parameters p, m, n and λ^2 may be quite large, they may not simultaneously be big. When errors of order up to n^{-2} are acceptable, the approximations of Lee (1971b) or Posten and Bargmann (1964) are suggested; for smaller n the exact powers are calculable.

Results correct to at least six significant figures have been calculated for various combinations of p up to 13, m up to 50, n up to 120 and λ^2 up to 32. With a lower degree of accuracy they can be extended further, but, for reasons previously mentioned, accurate results cannot be achieved with all parameters large. All calculations were performed using double precision arithmetic on a Univac 1106 at the University of Cape Town's Computer Centre. In accordance with the results of Anderson, Das Gupta and Mudholkar (1964) and of Das Gupta and Perlman (1973), powers of $|L|$ all increased as α, λ or n increased, and as m or p decreased. A selection of exact percentiles and powers is given in Tables 6, 7 and 8.

TABLE 1

Gupta's exact (G), Posten and Bargmann's approximation (PB) and Hart and Money's exact (HM) percentage points when $\alpha = 0,95$

p	m	n	λ^2	G	PB	HM
2	2	2	0,5	0,571752	0,572598	0,571752
			1,0	0,541292	0,543222	0,541292
			4,0	0,385821	0,398234	0,385821
2	2	8	0,5	0,898552	0,898554	0,898552
			1,0	0,886876	0,886881	0,886876
			4,0	0,802938	0,803031	0,802938
2	2	20	0,5	0,959648	0,959648	0,959648
			1,0	0,954656	0,954656	0,954656
			4,0	0,915845	0,915839	0,915836
3	2	8	0,5	0,775704	0,775729	0,775704
			1,0	0,760957	0,761017	0,760957
			4,0	0,670107	0,676552	0,670098
			8,0			0,559692
			16,0			0,405089
3	2	20	0,5	0,910887	0,910897	0,910897
			1,0	0,904028	0,904040	0,904039
			4,0	0,857452	0,857602	0,857585
			8,0			0,789103
			16,0			0,662312

TABLE 2

m = 2 n = 100 $\alpha = 0,05$

λ^2	P = 2		P = 3		P = 4	
	ITO	EXACT	ITO	EXACT	ITO	EXACT
1	,103	,1030	,092	,0912	,085	,0844
2	,165	,1658	,141	,1407	,126	,1256
3	,234	,2349	,196	,1964	,173	,1726
4	,305	,3073	,255	,2564	,224	,2240
5	,377	,3803	,317	,3189	,278	,2784
6	,447	,4517	,378	,3820	,333	,3346
7	,513	,5197	,439	,4442	,388	,3912
8	,576	,5832	,498	,5044	,443	,4471

TABLE 3

Ito's approximate (ITO), Mikhail's Exact (ME), Hart and Money's exact (HM) and Patnaik's approximate (PA) powers of $U_{2,2,97}$ when $\alpha = 0,05$

λ^2	ITO	ME	HM	PA
1	,103	,104	,1030	,105
2	,165	,168	,1656	,169
4	,305	,309	,3070	,312

TABLE 4

Mikhail's approximate (MA), Mikhail's adjusted (MB) and Hart and Money's exact (HM) powers of $U_{2,4,n}$ when $\alpha = 0,05$

λ^2	n = 45			n = 120		
	MA	HM	MB	MA	HM	MB
1.	,087	,0820	,081	,087	,0848	,085
2	,134	,1199	,118	,132	,1266	,126
4	,252	,2096	,208	,244	,2264	,227

TABLE 5

Roy's Gamma approximation (RG), de Waal's χ^2 approximation (DeW) and Hart and Money's exact (HM) powers for $U_{p,4,50}$ when $\alpha = 0,05$

λ^2	P = 3			P = 4		
	RG	DeW	HM	RG	DeW	HM
2	,1053	,1029	,1044	,0960	,0931	,0948
6	,2526	,2505	,2558	,2180	,2126	,2194
10	,4168	,4262	,4321	,3584	,3610	,3705

TABLE 6

Some Noncentral Percentiles of $|L|$

		P = 5		M = 6		N = 24		P = 6		M = 6		N = 20	
LAMBDA SQUARED	$\alpha =$	,100	,050	,025	,010	,005	,100	,050	,025	,010	,005	,100	,050
,000		,18397	,15854	,13842	,11727	,10423	,08522	,06980	,05821	,04666	,03989	,08522	,06980
,500		,18017	,15506	,13523	,11442	,10161	,08326	,06011	,05675	,04544	,03881	,08326	,06011
1,000		,17653	,15176	,13221	,11174	,09915	,08139	,06651	,05536	,04428	,03780	,08139	,06651
2,000		,16971	,14559	,12661	,10679	,09462	,07791	,06354	,05280	,04216	,03594	,07791	,06354
4,000		,15764	,13477	,11687	,09826	,08688	,07180	,05838	,04838	,03852	,03277	,07180	,05838
6,000		,14726	,12557	,10866	,09113	,08046	,06661	,05403	,04469	,03550	,03016	,06661	,05403
8,000		,13823	,11763	,10161	,08506	,07501	,06214	,05031	,04155	,03295	,02797	,06214	,05031

		P = 7		M = 6		N = 15		P = 8		M = 4		N = 25	
LAMBDA SQUARED	$\alpha =$	,100	,050	,025	,010	,005	,100	,050	,025	,010	,005	,100	,050
,000		,01788	,01314	,00993	,00706	,00555	,14665	,12445	,10715	,08926	,07840	,14665	,12445
,500		,01739	,01277	,00964	,00685	,00538	,14356	,12168	,10466	,08708	,07642	,14356	,12168
1,000		,01693	,01241	,00937	,00665	,00522	,14061	,11904	,10230	,08503	,07456	,14061	,11904
2,000		,01608	,01177	,00887	,00629	,00492	,13507	,11413	,09791	,08123	,07114	,13507	,11413
4,000		,01461	,01067	,00802	,00567	,00444	,12529	,10552	,09028	,07467	,06527	,12529	,10552
6,000		,01340	,00976	,00733	,00517	,00404	,11689	,09821	,08385	,06919	,06040	,11689	,09821
8,000		,01237	,00900	,00675	,00475	,00371	,10959	,09189	,07833	,06453	,05626	,10959	,09189

TABLE 7

Some Powers of $|L|$

LAMBDA SQUARED	P = 5 M = 6 N = 24					P = 7 M = 6 N = 15				
	$\alpha = .100$	.050	.025	.010	.005	.100	.050	.025	.010	.005
.500	.10947	.05564	.02824	.01149	.00582	.10599	.05350	.02697	.01089	.00548
1,000	.11923	.06156	.03167	.01311	.00671	.11204	.05706	.02899	.01181	.00598
2,000	.13956	.07418	.03916	.01670	.00872	.12432	.06440	.03320	.01375	.00704
4,000	.18303	.10240	.05656	.02543	.01377	.14942	.07980	.04223	.01802	.00941
6,000	.22934	.13417	.07708	.03629	.02027	.17506	.09606	.05202	.02279	.01210
8,000	.27749	.16893	.10054	.04931	.02832	.20101	.11301	.06250	.02803	.01512

LAMBDA SQUARED	P = 6 M = 6 N = 20					P = 8 M = 4 N = 25				
	$\alpha = .100$	.050	.025	.010	.005	.100	.050	.025	.010	.005
.500	.10782	.05463	.02763	.01120	.00565	.10873	.05518	.02796	.01136	.00574
1,000	.11582	.05942	.03039	.01248	.00636	.11770	.06059	.03108	.01282	.00654
2,000	.13229	.06948	.03628	.01526	.00790	.13632	.07205	.03784	.01603	.00834
4,000	.16691	.09141	.04951	.02173	.01157	.17590	.09746	.05335	.02372	.01274
6,000	.20325	.11548	.06459	.02942	.01606	.21792	.12584	.07143	.03312	.01831
8,000	.24074	.14139	.08139	.03833	.02140	.26157	.15676	.09193	.04427	.02510

TABLE 8

Some Powers of $|L|$

LAMBDA SQUARED	P = 7 M = 2 N = 60					P = 8 M = 2 N = 40				
	$\alpha = .100$	.050	.025	.010	.005	.100	.050	.025	.010	.005
.500	.11752	.06078	.03134	.01302	.00669	.11498	.05909	.03029	.01248	.00637
1,000	.13608	.07256	.03848	.01654	.00870	.13072	.06890	.03613	.01530	.00796
2,000	.17593	.09902	.05518	.02517	.01380	.16427	.09063	.04951	.02202	.01184
4,000	.26384	.16239	.09816	.04939	.02899	.23788	.14177	.08301	.04007	.02280
6,000	.35755	.23672	.15300	.08347	.05184	.31690	.20149	.12506	.06469	.03860
8,000	.45144	.31789	.21757	.12727	.08300	.39760	.26735	.17460	.09597	.05971

LAMBDA SQUARED	P = 9 M = 4 N = 15					P = 13 M = 4 N = 15				
	$\alpha = .100$	.050	.025	.010	.005	.100	.050	.025	.010	.005
.500	.10580	.05335	.02687	.01084	.00545	.10291	.05163	.02588	.01038	.00520
1,000	.11164	.05677	.02879	.01170	.00591	.10580	.05325	.02677	.01076	.00540
2,000	.12344	.06374	.03275	.01351	.00689	.11153	.05650	.02854	.01153	.00580
4,000	.14742	.07826	.04116	.01742	.00904	.12281	.06294	.03209	.01309	.00662
6,000	.17171	.09343	.05017	.02173	.01144	.13385	.06934	.03565	.01467	.00746
8,000	.19614	.10911	.05969	.02639	.01408	.14465	.07568	.03921	.01626	.00831

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